



ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY

DEPARTMENT OF MATHEMATICS

APPLIED MATHEMATICS

For freshman engineering regular students

DATE: Dec 22, 2021

APPLIED MATHEMATICS-I

TEST- II | 10% |

TIME ALLOWED: 45 MINUTES

NAME: _____ IDNO: _____ Section: _____

Part-I Short answer | 1.5 pts each |

1. If $\int_4^{10} f(x)dx = 4.6$ and $\int_1^{10} f(x)dx = 12.5$ then $\int_1^4 f(x)dx =$ _____

2. Evaluate: $\int_{-2}^3 \frac{1}{x^3} dx =$ _____

3. $\int \frac{10-3x}{x^2-4} dx =$ _____

4. If $F(x) = \int_0^{x^6} e^{t^2} dt$, then $F'(x) =$ _____

Part-II Work out

Solve the following questions with the necessary steps clearly. [2 pts each]

1. Find the area of the region enclosed by the parabolas $y = x^2$ and $y = 2x - x^2$

2. Evaluate: $\int \sqrt{4x - x^2} dx$